WARDA SHAHZAD

PERSONAL DETAILS

Abbottabad
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Nationality: Pakistan
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Google Scholar: scholar.google.com/citations?user=Y5FmMc8AAAAJ&hl=en
ResearchGate: researchgate.net/profile/Warda-Shahzad

PROFILE

I study how small changes in composition and processing reshape the structure and optical behavior of semiconductor materials. My doctoral research focuses on Li, Na, and K modified CuZnSnS (CZTS) thin films, with particular interest in defect engineering, crystallinity, and bandgap control.
I'm a PhD researcher in Chemistry (thesis submitted, external viva pending) at COMSATS University Islamabad, Abbottabad Campus, working on the structural, optical, and electronic properties of kesterite thin films for thin-film solar cells. My research centers on alkali metal doping with lithium, sodium, and potassium, targeting suppression of Cu/Zn antisite defects and secondary phases (ZnS, CuSnS), the primary loss mechanisms limiting CZTS device efficiency.
I synthesize films by solution-based chemical deposition and characterize them across XRD, SEM/EDX, and UV-Vis spectroscopy, using dislocation density and microstrain as quantitative proxies for defect suppression across each doping series. A DFT chapter in my thesis complements this experimental work, computing structural optimization, density of states, and band structure to confirm CZTS's direct bandgap nature. I'm currently seeking a postdoctoral position to extend this work to new synthesis platforms and semiconductor material systems.

EDUCATION

PhD, Chemistry — 2020 - 2026
COMSATS University Islamabad, Abbottabad Campus, Abbottabad

- Thesis: “Exploration of the Physical Properties of Kesterite Thin Films Modified by Alkali Metal Cations for Sunlight Harvesting,” including Li/Na and Li/K co-doped systems and a DFT chapter on structural and electronic properties.
- Thesis submitted for foreign evaluation, external viva pending.

MS, Chemistry — 2017 - 2019
COMSATS University Islamabad, Abbottabad Campus, Abbottabad

- Thesis on visible-light photocatalysis using alkaline earth zirconate perovskite/graphene oxide composites, published in Ceramics International (2022).

MSc, Chemistry — 2015 - 2017
University of Wah, Wah

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SKILLS

Chemical bath deposition
Solution-based thin-film synthesis
Two-step thermal annealing
Composition screening
XRD structural analysis
Lattice parameters analysis
Crystallite size analysis
Microstrain analysis
Dislocation density analysis
Williamson-Hall analysis
VESTA structure models
SEM morphological analysis
EDS/EDX elemental composition analysis
Grain size analysis
UV-Vis spectroscopy
Tauc analysis
Refractive index determination
Dielectric response measurement
Optical and electrical conductivity measurement
DFT structural optimization
Density of states calculations
Band structure calculations

COURSES

Advanced analytical instrumentation 2026
National Centre for Physics, Islamabad.

REFERENCES

Prof. Dr. Bushra Ismail
COMSATS University Islamabad, Abbottabad Campus, Abbottabad
bushraismail@cuiatd.edu.pk

Prof. Dr. Asad Muhammad Khan
COMSATS University Islamabad, Abbottabad Campus, Abbottabad
amk@cuiatd.edu.pk

Prof. Dr. Rafaqat Ali Khan
COMSATS University Islamabad, Abbottabad Campus, Abbottabad
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ACHIEVEMENTS

- 3 first-author papers during PhD.
- 2nd Prize, Poster Presentation, SSCS, Department of Chemistry, COMSATS University Islamabad, Abbottabad Campus, 2025.
- Reviewer, Journal of Materials Science: Materials in Electronics (Springer Nature), 2025.

PUBLICATIONS

First-Author, PhD Research
- Optical Materials (2025): Li-doped CZTS.
- Journal of Physics and Chemistry of Solids (2025): Na-doped CZTS.
- K-doped CZTS (Submitted).

First-Author, MS Research

- Ceramics International (2022).

Co-Authored & Peer Review

- Materials Science and Engineering B, 2024.
- Journal of Inorganic and Organometallic Polymers, 2025.
- Reviewer, Journal of Materials Science: Materials in Electronics, 2025.

MENTORING & RESEARCH COMMUNICATION

- Mentored MS students in materials chemistry synthesis and characterization throughout doctoral studies.
- Oral and poster presentations at ICRA-C 2024, NUST Islamabad, and ICC RTC&RET 2023, University of Swat.